INCH-POUND

MIL-C-70549 (AR)
AMENDMENT 4
14 July 1994
SUPERSEDING
AMENDMENT 3
12 May 1993

MILITARY SPECIFICATION

CARTRIDGE, 81MM, SMOKE, RED PHOSPHORUS
M819, WITH FUZE, MTSQ, M772
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of MIL-F-70549 (AR), dated 5 September 1986, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1.2, Heading: Delete "DRAWINGS", and substitute "PRODUCT AND PACKAGING DRAWINGS".

Under PRODUCT AND PACKAGING DRAWINGS, delete:

"8796521 Marking ... Containers
8796522 Marking ... Boxes".

Add:

"12944509 Packing and Marking for Container, Ammo, PA156 and PA157; for Cartridge, 81mm".

After PRODUCT AND PACKAGING DRAWINGS, add:

"INSPECTION EQUIPMENT DRAWINGS
6518071 Alignment Gage
6518072 Alignment Check Gage".

AMSMC N/A
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PAGE 4

3.4.6, Delete in its entirety and substitute:

"3.4.6 Fuze function: The fuze shall air burst when set in accordance with paragraph 4.5.5, Table III."

PAGE 5

* 4.3.1, Delete: "Body tube assembly 9327832 5".

Change "9362899" to "9362799".

Add the following under "Projectile loading assembly ...9327838 ...5".

"Projectile loading assembly (when LAP is split between two facilities) ...9327838 ...50*".

In last line under "Quantity" delete "5" and substitute "50*".

After list add "*Same group of samples when LAP is split".

PAGE 6

* TABLE I: Add "4.4.2.2" before "4.4.2.5" in "Paragraph Reference" column.

In "Projectile Loading Assembly ..." add "1/" next to "5" in "No. of sample units" column.

Under "Projectile Loading Assembly ...Examination for defects...": add: "Ballistic test ...50 2/ ...100% ...3.2 ...4.5.5".

In "Cartridge, 81MM Smoke ...Examination for defects ...": Change "5" to "50" in "No. of sample units" column.

In "Ballistic test ..." add "3/" next to "50" in "No. of sample units" column.

Add Notes "1/ Sample shall be 50 when LAP is split
2/ Required only when LAP is split
3/ Same units as in Note 2/ when LAP is split".

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Change "9354333" to "9362799" where it appears.

Change "9362899" to "9362799" and change ballistic test sample quantity from "5" to "50".

PAGE 8

* 4.4.2.1: Delete in its entirety.

PAGE 9

4.4.2.2., Major 105: Change to read, "Piston plate assembly missing or foil punctured".

Add Major 110, 111, 112 and 113 as follows:

"110 ...Presence of foreign matter between foil disc and piston plate or presence of contaminant on piston plate ...0.40% ...3.2 ...Visual/Manual"

111 ...Foil punctured, wrinkled or not adhering fully to piston plate ...0.40% ...3.2 ...Visual

112 ...Concentricity of foil disc with piston plate outside diameter (OD) ...0.40% ...3.2 ...Gage

* 113 ...Presence of foreign matter in tail cone groove ...0.40% ...3.2 ...Visual".

PAGE 10

4.4.2.3, Major 105: delete "5" in "No. of sample units" column and delete "100%" in "AQL or 100%" column.

Delete Note "1/..." in its entirety and substitute "1/ ...See 4.4.3.1".

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4.4.2.5, Major 109: Change "0.40%" to "100%", and change "Visual" to "Gage".

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4.4.2.5, Delete "Major 112" in its entirety.

Add new "Major 112 ...Assembly fails to freely traverse alignment gage ...1/ ...3.2 ...Gage".

Add Note "1/. Applies to split LAP only. At start-up, 30 consecutive assemblies must successfully pass. Thereafter, one out of each 20 assemblies shall be checked. If any failures occur, the preceding 50 assemblies shall be checked. If any more failures occur, continue to screen groups of 50 until all pass."

PAGE 15

In Next Higher Assembly block change "9362899" to "9362799".

4.4.2.7, Major 106: Delete in its entirety and substitute:

"106 Inadequate torque applied to fin boom or fuze ...1/ ...(-) ...3.2 ...Torque test/4.5.2".

Note 1/: add "and 5 assemblies per hour thereafter. Any failures shall cause rejection back to the last successful inspection."

PAGE 16

Delete "4.4.2.8" in paragraph block and substitute "4.4.2.7".

In Next Higher Assembly block change "9362899" to "9362799".

Major 109, delete "100%" in "AQL" column.

Note 1/, delete second sentence in its entirety and substitute:

"Repaint and recheck if the ring inspection exposes bare metal."

Major 110, delete in its entirety.

Major 111, delete "Foil tape not centered on back of fuze" and substitute:

"Foil tape touches any part of fuze threads".

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4.4.2.9, In drawing number block, delete "9354333" and substitute "9362799".

In next higher assembly block, delete "9362799".

* In next higher assembly block, add "/12944509" after "9362798".

Under major category, add the following:

"104 Propelling charge support missing or damaged ...0.40% ...3.2 ...Visual

105 Increment tape missing ...0.40% ...3.2 ...Visual

106 Tab at the end of increment tape inadequate ...0.40% ...3.2 ... Visual

107 Tab at the end of increment tape covered by propellant charge support label ...0.40% ...3.2 ...Visual".

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* 4.4.2.10, In Drawing Number block, change "9362899" to "9362799".

* Drawing number block, delete: "/8796521".

* Next higher assembly block, delete: "9354616" and substitute "9362798/12944509".

* Add Major: "105 ...Tab too small ...- ...0.40% ...3.2 ...Gage".

PAGE 19

* Change Title block to, "Box, Packing for Container, Ammunition Fiber for Cartridges, 81MM, M819/M853, prior to closing".

* Drawing number block, delete: "9354616" and substitute "9362798".

* Next higher assembly block, delete: "9362798/8796522".

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4.4.2.11, Major 102: delete in its entirety.

Add Minor 203 and 204 as follows:

"203 Plastic sheet not present in every sixth box ...(-) ...0.65% ...3.2 ...Visual

204 Box containing plastic sheet not appropriately marked ...(-) ...0.65% ...3.2 ...Visual".

Add new minor defects 205 and 206 as follows:

"205 Graphical firing scale not present in every third box ...(-) ...0.65% ...3.2 ...Visual

206 Box containing graphical firing scale not appropriately marked ...(-) ...0.65% ...3.2 ...Visual".

PAGE 20

In Title block, change "Wirebound" to "Wooden" where it appears.

PAGE 20a

* Add new paragraph 4.4.2.13 (Classification of Defects and Tests):

"4.4.2.13 ...Packing and Marking for Container, Ammo, PA156 and PA157 for Cartridge, 81mm (prior to closing) ...sheet 1 of 1 ...drawing number 12944509

CRITICAL None defined.

MAJOR

101 Containers loose or improperly packed in metal can ...(-) ...0.40% ...3.2 ...Visual/Manual

102 Support assembly damaged, missing, or improperly placed ...(-) ...0.40% ...3.2 ...Visual

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MINOR

201 Circular firing scale not present in every third metal box ...(-) ...0.65% ...3.2 ...Visual

202 Ammo box containing circular firing scale not appropriately marked ...(-) ...0.65% ...3.2 ...Visual

203 Plastic sheet not present in every sixth box ...(-) ...0.65% ...3.2 ...Visual

204 Ammo box containing plastic sheet not appropriately marked ...(-) ...0.65% ...3.2 ...Visual

205 Evidence of poor workmanship ...(-) ...0.65% ...3.2 ...Visual".

PAGE 20b

* Add new paragraph 4.4.2.14 (Classification of Defects and Tests):

"4.4.2.14 ...Packing and Marking for Container, Ammo, PA156 and PA157 for Cartridge, 81mm (after closing) ...sheet 1 of 1 ...drawing number 12944509

MAJOR

101 Contents of box loose ...(-) ...0.40% ...3.2 ...Manual

102 Cover assembly not secure ...(-) ...0.40% ...3.2 ...Manual

MINOR

201 Marking missing, misleading or unidentifiable ...(-) ...0.65% ...3.2 ...Visual

202 Evidence of poor workmanship ...(-) ...0.65% ...3.2 ...Visual".

PAGE 21

Under "Precaution..." add new paragraph 4.4.3.1 as follows:

"4.4.3.1 Tack weld pull test. Ten consecutive assemblies shall pass this test whenever equipment is started. Five (5) shall be tested each hour thereafter and ten (10) whenever tooling is changed or equipment is adjusted. Failure of any sample shall cause rejection of all obturators welded since the last successful test. See 4.5.4."

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4.4.5, Table II: Delete "Fuze non-function ...2" and substitute:

"Fuze non-functions, functions upon ground impact or fails to separate tail cone from body tube assembly ...1 ...2 (See Note 1/)".

Delete: "Major ...Fuze functioning time (See Note 1/) ...2 ...3"

Delete "Note 1/: ..." in its entirety and substitute:

"Note 1/: If two (2) failures occur, another sample of 50 shall be fired. If two (2) or more additional failures occur the lot shall be rejected. If three (3) or more failures occur during the first 50 fired, the lot shall be rejected."

PAGE 22

4.5.2, Change "4.2.3.4" to "4.4.2.7"

Add to the end of the last sentence:

"and shall cause rejection of all cartridges produced since the last successful test."

4.5.5, Delete full sentences 5, 6 and 7 ("However ..." "...meters") in their entirety and substitute:

"Observation shall be made for compliance with the requirements specified for security of parts, air burst, horizontal range and smoke production. Horizontal range shall be reported to the point on the ground under the position of fuze functioning and shall be measured to the nearest five (5) meters."

Delete:

"The cartridges shall be conditioned at 70 plus or minus 2.5 degrees Fahrenheit for a minimum of 4 hours prior to testing and fired at this temperature." and substitute:

"The cartridges shall be conditioned at 70 plus or minus 2.5 degrees Fahrenheit for a minimum of 24 hours prior to testing and shall be fired at this temperature."

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Delete "TABLE III..." in its entirety and substitute:

"TABLE III

Fuze Setting (sec.)	Charge Zone	Elevation (mils)
24.5	1	1459
46.0	4	1489

PAGE 23

* 4.5.5, After the last sentence, add:

"The cannon shall be dry swabbed after every tenth round fired, regardless of charge."

* 5.1, Delete in its entirety and substitute:

"5.1 Unit packaging.

5.1.1 Level A. Packaging for cartridges shall be in accordance with the requirements of drawing 9362799. Prior to taping, the felt spacers shall cause a gap of approximately 1/8 inch between the cap and tube of the fiber container when unrestrained.

5.1.2 Level B. Same as Level A.

5.1.3 Level C. Not applicable."

* Add new paragraph:

"5.2 Packing.

5.2.1 Level A. Cartridges shall be packed in accordance with the requirements of drawing 9362798 or 12944509, as applicable.

5.2.2 Level B. Same as Level A.

5.2.3 Level C. Not applicable."

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* Add new paragraph:

"5.3 Marking.

5.3.1 Level A. Packaged cartridges shall be marked in accordance with the requirements of drawing 9362799. Packed cartridges shall be marked with the requirements of drawing 9362798-1 or 12944509, as applicable.

5.3.2 Level B. Same as Level A.

5.3.3 Level C. Not applicable."

PAGE 24

* 6.4, Delete: "AMSMC-QAT-I (D), Dover, NJ 07801-5001" and substitute:

"SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000".

* 6.5, Delete: "AMSMC-QAT-B (D)/SMCAR-AEE-WW (D), Dover, NJ 07801-5000" and substitute:

"SMCAR-QAT-B/SMCAR-FSA-MM, Picatinny Arsenal, NJ 07806-5000".

* 6.6, Delete in its entirety and substitute:

"6.6 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool control inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when he provides, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to, or better than, the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provides equal assurance, the provisions of this specification shall apply."

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On Form 1426, change "M19" to "M819" where it appears.

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